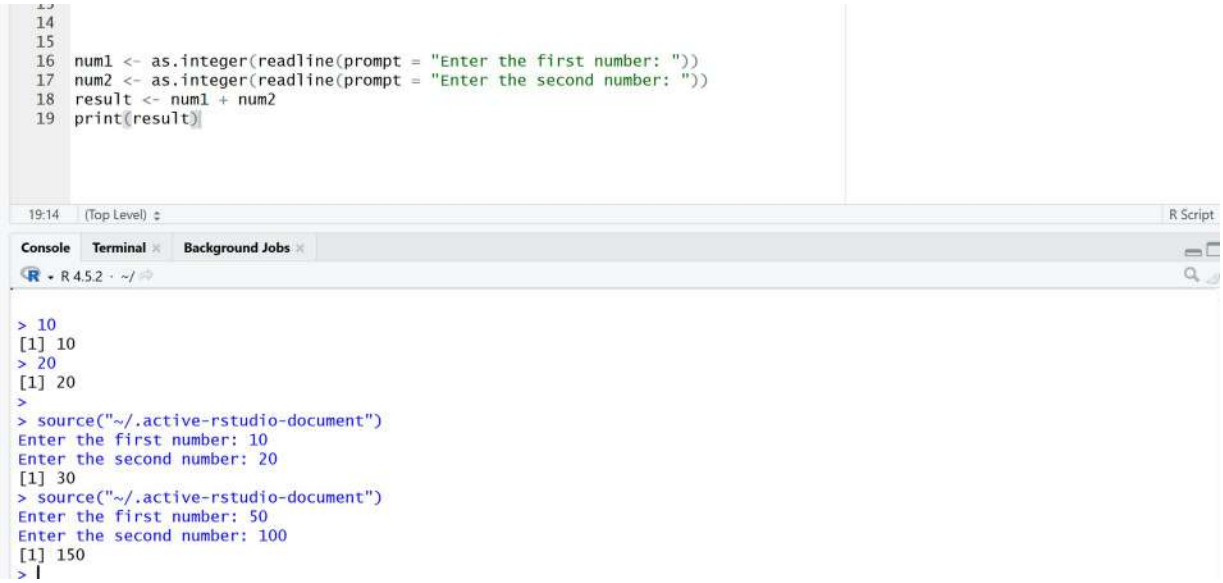
# Lab Experiments

## CSA-1614-DWDM

Rohith Manikanta Rellu
-192425326

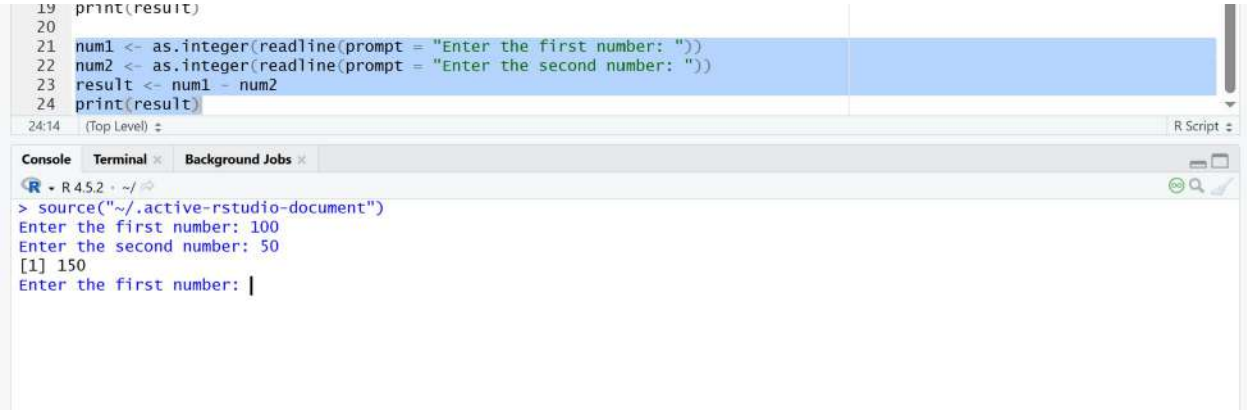
## 1. Addition of Two Numbers in R

### Output



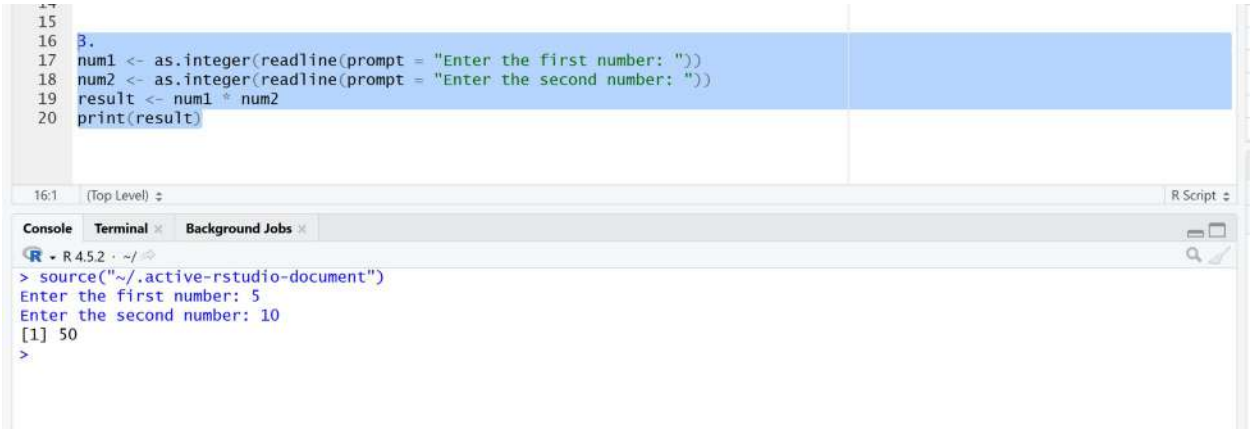

## 2. Subtraction of Two Numbers in R

### Output



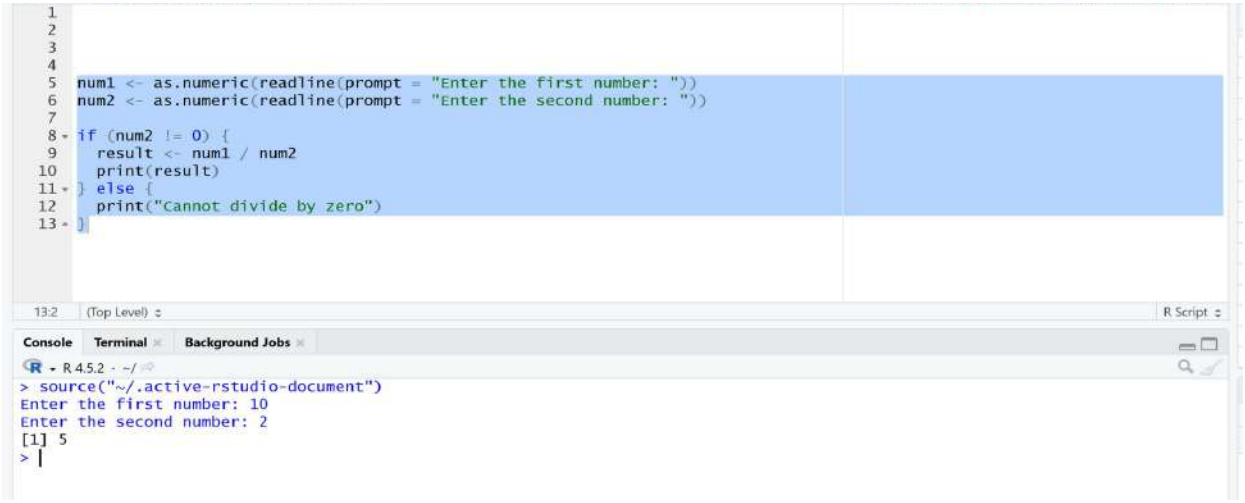
## 3. Multiplication of Two Numbers in R

### Output



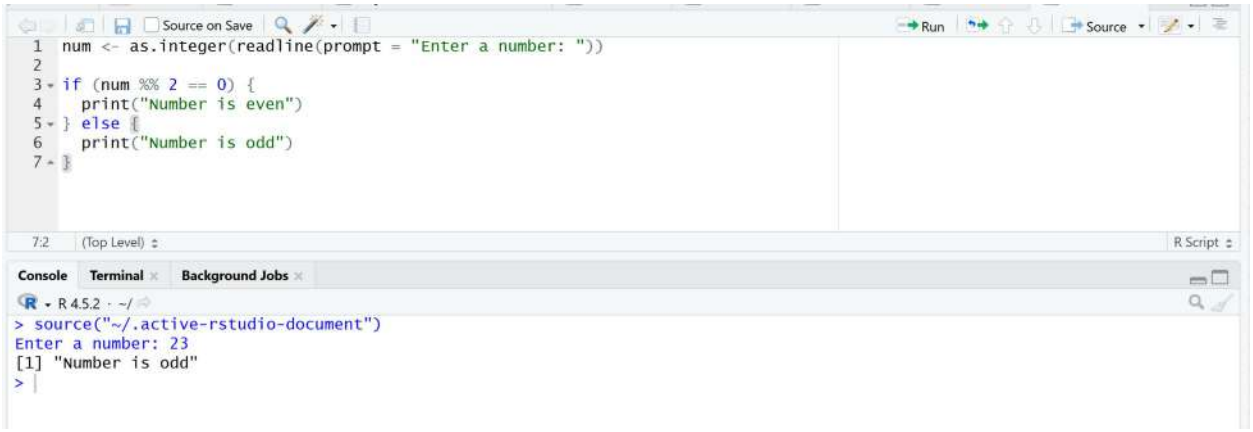

## 4. Division of Two Numbers in R

### Output



## 5. Checking Whether a Number Is Odd or Even

### Output



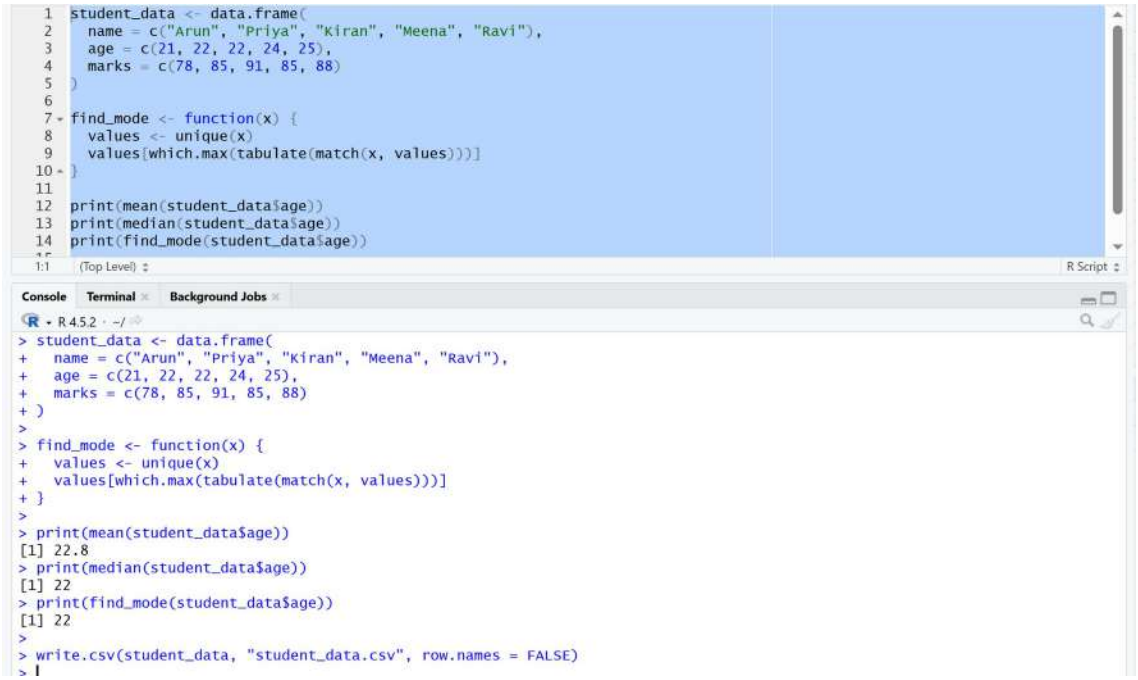
```
num <- as.integer(readline(prompt = "Enter a number: "))

if (num %% 2 == 0) {
  print("Number is even")
} else {
  print("Number is odd")
}
```

```
> source("~/.active-rstudio-document")
Enter a number: 23
[1] "Number is odd"
>
```

## 6. Calculating Mean, Median, and Mode

### Output



```
student_data <- data.frame(
  name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
  age = c(21, 22, 22, 24, 25),
  marks = c(78, 85, 91, 85, 88)
)

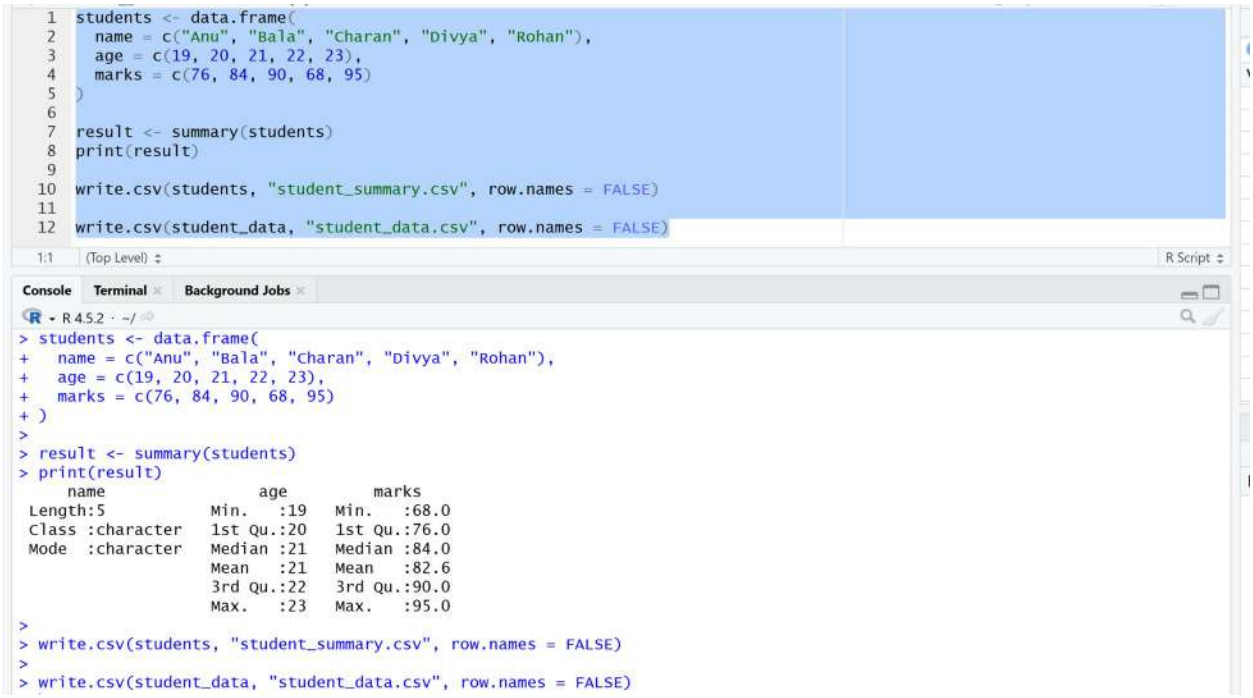
find_mode <- function(x) {
  values <- unique(x)
  values[which.max(tabulate(match(x, values)))]
}

print(mean(student_data$age))
print(median(student_data$age))
print(find_mode(student_data$age))
```

```
> student_data <- data.frame(
+   name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
+   age = c(21, 22, 22, 24, 25),
+   marks = c(78, 85, 91, 85, 88)
+ )
>
> find_mode <- function(x) {
+   values <- unique(x)
+   values[which.max(tabulate(match(x, values)))]
+ }
>
> print(mean(student_data$age))
[1] 22.8
> print(median(student_data$age))
[1] 22
> print(find_mode(student_data$age))
[1] 22
>
> write.csv(student_data, "student_data.csv", row.names = FALSE)
>
```

## 7. Generating a Statistical Summary of Data

### Output



```
students <- data.frame(
  name = c("Anu", "Bala", "Charan", "Divya", "Rohan"),
  age = c(19, 20, 21, 22, 23),
  marks = c(76, 84, 90, 68, 95)
)

result <- summary(students)
print(result)

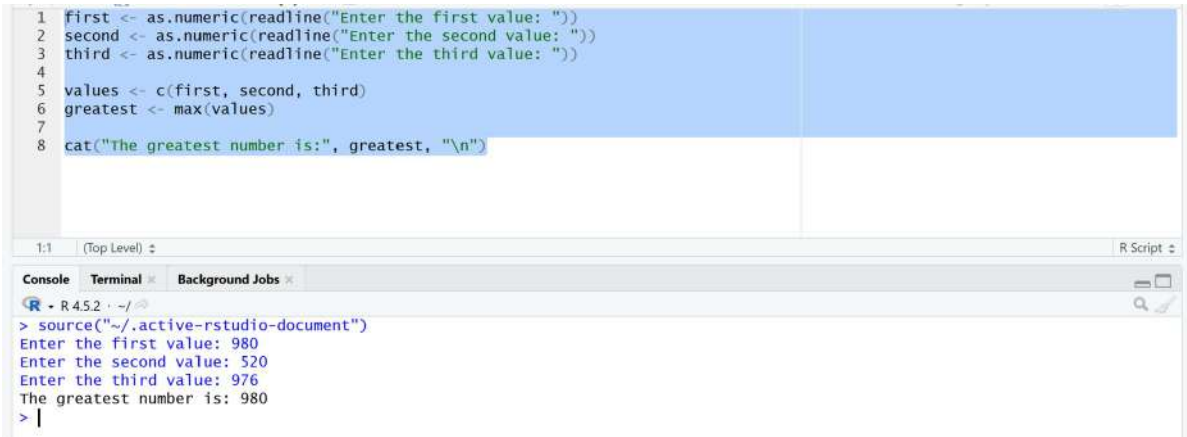
write.csv(students, "student_summary.csv", row.names = FALSE)

write.csv(student_data, "student_data.csv", row.names = FALSE)
```

```
> students <- data.frame(
+   name = c("Anu", "Bala", "Charan", "Divya", "Rohan"),
+   age = c(19, 20, 21, 22, 23),
+   marks = c(76, 84, 90, 68, 95)
+ )
>
> result <- summary(students)
> print(result)
     name                age        marks
 Length:5           Min.   :19   Min.   :68.0
 Class :character   1st Qu.:20   1st Qu.:76.0
 Mode  :character   Median :21   Median :84.0
                    Mean   :21   Mean   :82.6
                    3rd Qu.:22   3rd Qu.:90.0
                    Max.   :23   Max.   :95.0
>
> write.csv(students, "student_summary.csv", row.names = FALSE)
>
> write.csv(student_data, "student_data.csv", row.names = FALSE)
```

## 8. Finding the Greatest Among Three Numbers

### Output



```
first <- as.numeric(readline("Enter the first value: "))
second <- as.numeric(readline("Enter the second value: "))
third <- as.numeric(readline("Enter the third value: "))

values <- c(first, second, third)
greatest <- max(values)

cat("The greatest number is:", greatest, "\n")
```

```
> source("~/.active-rstudio-document")
Enter the first value: 980
Enter the second value: 520
Enter the third value: 976
The greatest number is: 980
>
```

## 9. Calculating the Interquartile Range (IQR)

### Output



```
student_data <- data.frame(
  name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
  age = c(20, 21, 22, 23, 24),
  marks = c(55, 67, 78, 89, 95)
)

result <- IQR(student_data$marks)
print(result)

write.csv(student_data, "iqr_data.csv", row.names = FALSE)
```

```
> student_data <- data.frame(
+   name = c("Arun", "Priya", "Kiran", "Meena", "Ravi"),
+   age = c(20, 21, 22, 23, 24),
+   marks = c(55, 67, 78, 89, 95)
+ )
>
> result <- IQR(student_data$marks)
> print(result)
[1] 22
>
> write.csv(student_data, "iqr_data.csv", row.names = FALSE)
```

## 10. Calculating Quantiles of Data

### Output



```
student_data <- data.frame(
  name = c("Asha", "Bala", "Charan", "Divya", "Rohan"),
  age = c(19, 20, 21, 22, 23),
  marks = c(65, 72, 81, 88, 96)
)

result <- quantile(student_data$marks, probs = c(0, 0.25, 0.5, 0.75, 1))
print(result)

write.csv(student_data, "quantile_data.csv", row.names = FALSE)
```

```
> student_data <- data.frame(
+   name = c("Asha", "Bala", "Charan", "Divya", "Rohan"),
+   age = c(19, 20, 21, 22, 23),
+   marks = c(65, 72, 81, 88, 96)
+ )
>
> result <- quantile(student_data$marks, probs = c(0, 0.25, 0.5, 0.75, 1))
> print(result)
  0%  25%  50%  75% 100%
  65   72   81   88   96
>
> write.csv(student_data, "quantile_data.csv", row.names = FALSE)
```

## 11. Calculating the Mid-Range of Data

### Output

```
scores <- c(45, 60, 72, 85, 95)

minimum <- min(scores)
maximum <- max(scores)
mid_range <- (minimum + maximum) / 2

print(minimum)
print(maximum)
print(mid_range)
```

```
> scores <- c(45, 60, 72, 85, 95)
>
> minimum <- min(scores)
> maximum <- max(scores)
> mid_range <- (minimum + maximum) / 2
>
> print(minimum)
[1] 45
> print(maximum)
[1] 95
> print(mid_range)
[1] 70
>
```

## 12. Performing Z-Score Normalization

### Output

```
marks <- c(55, 65, 70, 80, 90, 95)

average <- mean(marks)
std_dev <- sd(marks)

z_scores <- (marks - average) / std_dev

print(z_scores)
```

```
> marks <- c(55, 65, 70, 80, 90, 95)
>
> average <- mean(marks)
> std_dev <- sd(marks)
>
> z_scores <- (marks - average) / std_dev
>
> print(z_scores)
[1] -1.3614329 -0.7079451 -0.3812012  0.2722866  0.9257743  1.2525182
>
```

## 13. Calculating Minimum, Maximum, Mean, and Min-Max Normalization

### Output



```
values <- c(10, 20, 30, 40, 50, 60)
average <- mean(values)
minimum <- min(values)
maximum <- max(values)

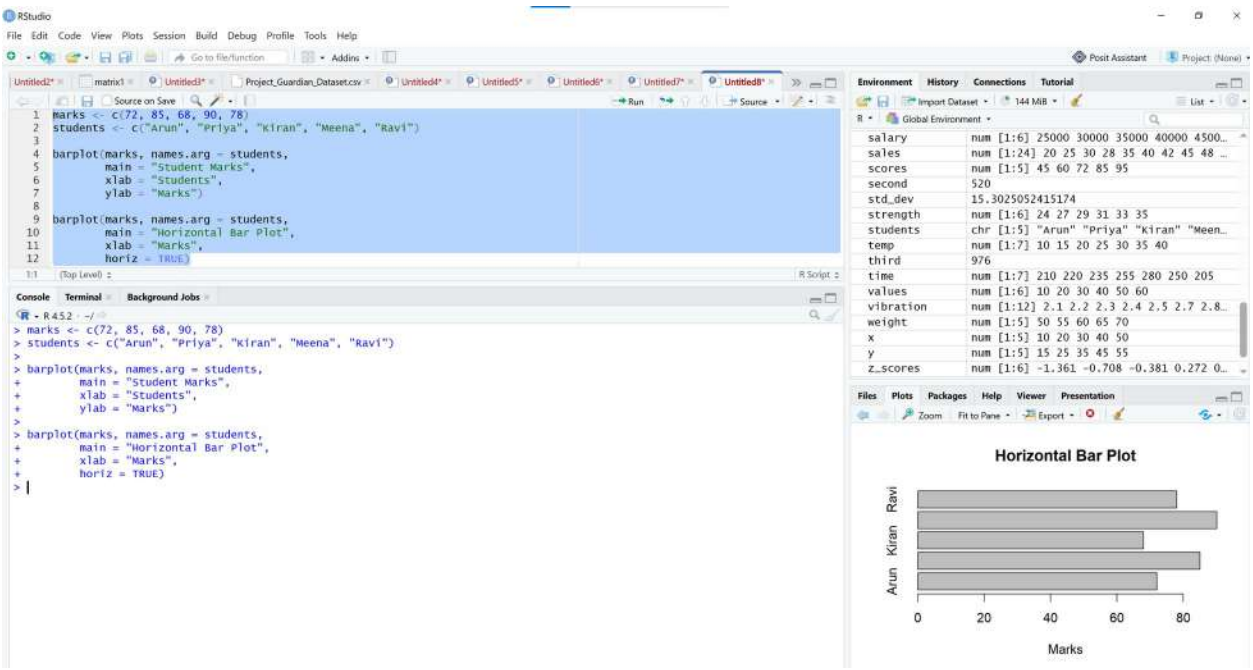
normalized <- (values - minimum) / (maximum - minimum)

print(average)
print(minimum)
print(maximum)
print(normalized)
```

```
> values <- c(10, 20, 30, 40, 50, 60)
> average <- mean(values)
> minimum <- min(values)
> maximum <- max(values)
>
> normalized <- (values - minimum) / (maximum - minimum)
>
> print(average)
[1] 35
> print(minimum)
[1] 10
> print(maximum)
[1] 60
> print(normalized)
[1] 0.0 0.2 0.4 0.6 0.8 1.0
```

## 14. Drawing Vertical and Horizontal Bar Plots

### Output



```
marks <- c(72, 85, 68, 90, 78)
students <- c("Arun", "Priya", "Kiran", "Meena", "Ravi")

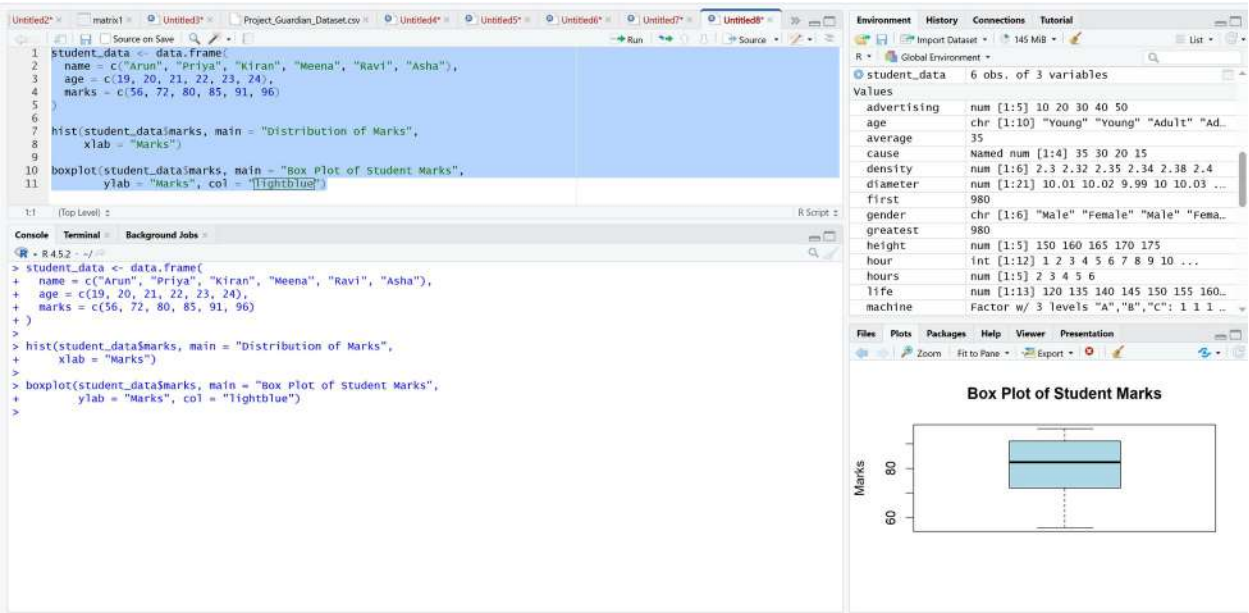
barplot(marks, names.arg = students,
        main = "Student Marks",
        xlab = "Students",
        ylab = "Marks")

barplot(marks, names.arg = students,
        main = "Horizontal Bar Plot",
        xlab = "Marks",
        horiz = TRUE)
```

```
> marks <- c(72, 85, 68, 90, 78)
> students <- c("Arun", "Priya", "Kiran", "Meena", "Ravi")
>
> barplot(marks, names.arg = students,
+         main = "Student Marks",
+         xlab = "Students",
+         ylab = "Marks")
>
> barplot(marks, names.arg = students,
+         main = "Horizontal Bar Plot",
+         xlab = "Marks",
+         horiz = TRUE)
>
```

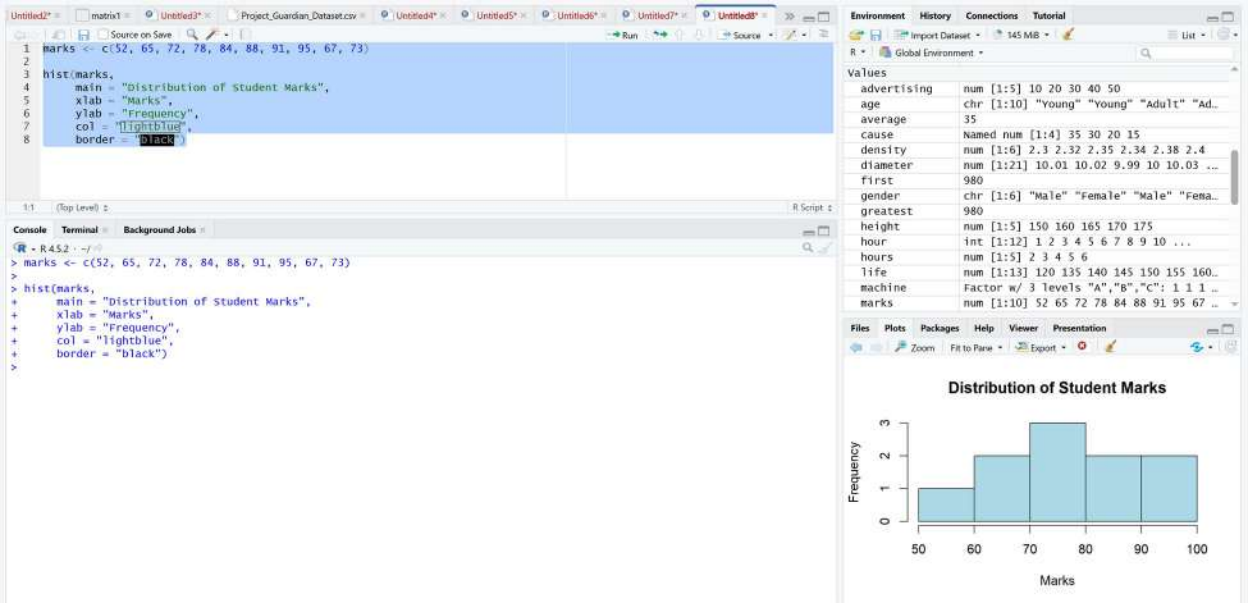
## 15. Visualizing Data Using a Box Plot

### Output



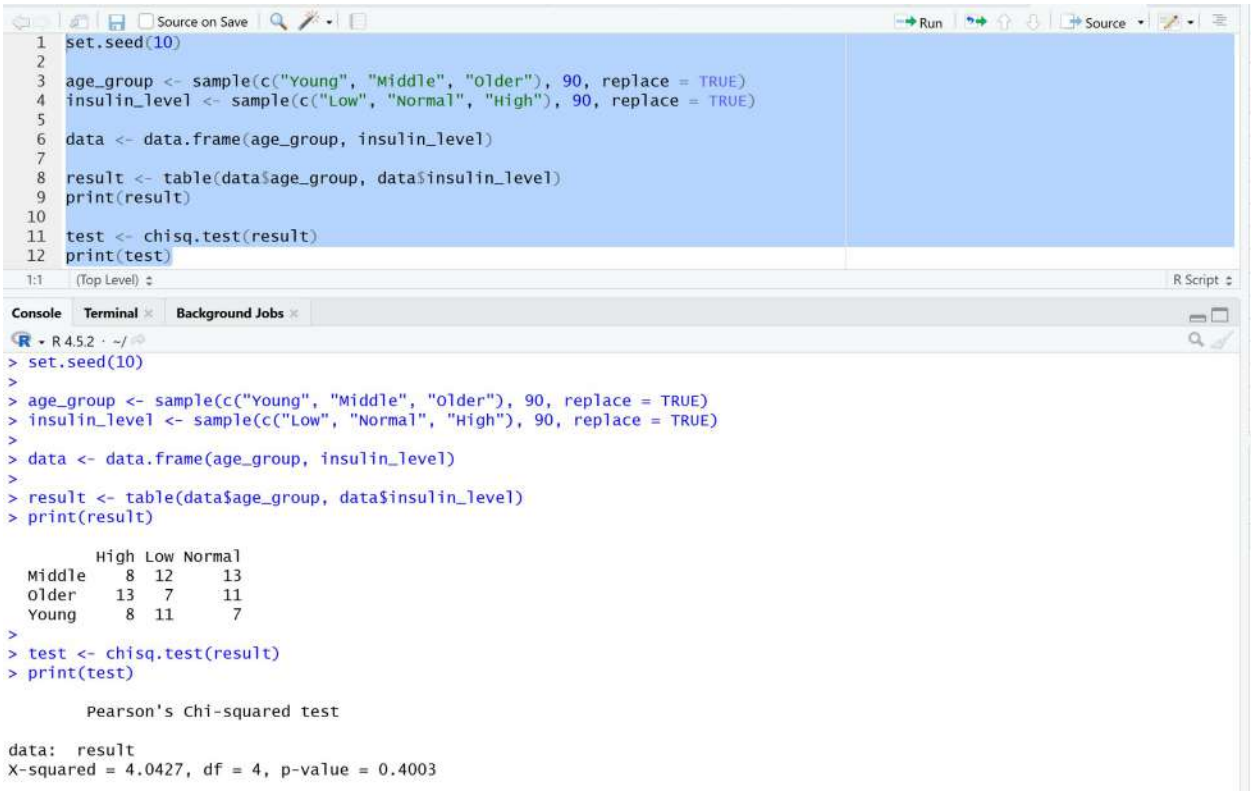

## 16. Creating a Histogram in R

### Output



## 17. Performing Chi-Square Analysis of Data

### Output



```
set.seed(10)

age_group <- sample(c("Young", "Middle", "Older"), 90, replace = TRUE)
insulin_level <- sample(c("Low", "Normal", "High"), 90, replace = TRUE)

data <- data.frame(age_group, insulin_level)

result <- table(data$age_group, data$insulin_level)
print(result)

test <- chisq.test(result)
print(test)
```

```
> set.seed(10)
>
> age_group <- sample(c("Young", "Middle", "Older"), 90, replace = TRUE)
> insulin_level <- sample(c("Low", "Normal", "High"), 90, replace = TRUE)
>
> data <- data.frame(age_group, insulin_level)
>
> result <- table(data$age_group, data$insulin_level)
> print(result)

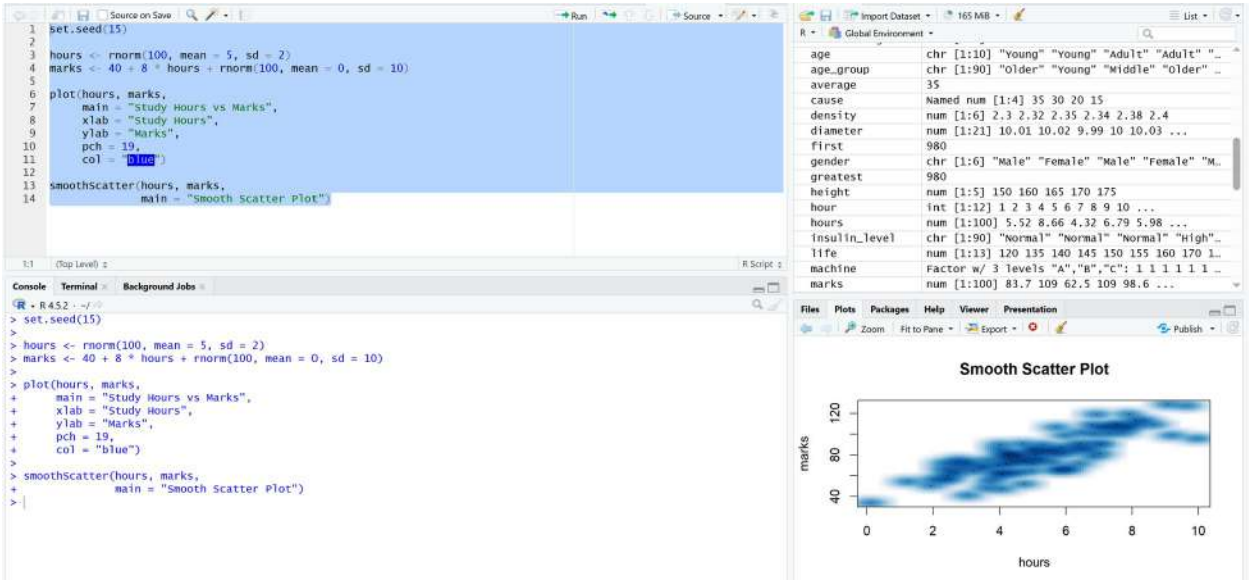
         High Low Normal
  Middle    8  12     13
  Older    13   7     11
  Young     8  11      7
>
> test <- chisq.test(result)
> print(test)

        Pearson's Chi-squared test

data:  result
X-squared = 4.0427, df = 4, p-value = 0.4003
```

## 18. Creating a Scatter Plot in R

### Output



```
set.seed(15)

hours <- rnorm(100, mean = 5, sd = 2)
marks <- 40 + 8 * hours + rnorm(100, mean = 0, sd = 10)

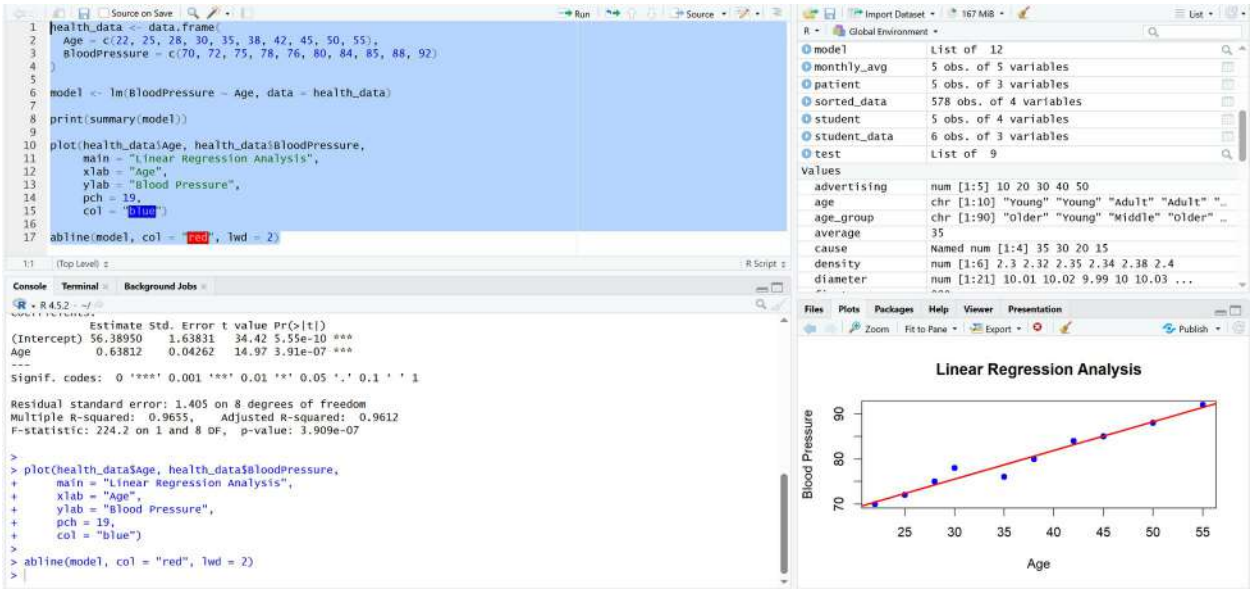
plot(hours, marks,
     main = "Study Hours vs Marks",
     xlab = "Study Hours",
     ylab = "Marks",
     pch = 19,
     col = "blue")

smoothScatter(hours, marks,
              main = "Smooth Scatter Plot")
```

```
> set.seed(15)
>
> hours <- rnorm(100, mean = 5, sd = 2)
> marks <- 40 + 8 * hours + rnorm(100, mean = 0, sd = 10)
>
> plot(hours, marks,
+      main = "Study Hours vs Marks",
+      xlab = "Study Hours",
+      ylab = "Marks",
+      pch = 19,
+      col = "blue")
>
> smoothScatter(hours, marks,
+               main = "Smooth Scatter Plot")
>
```

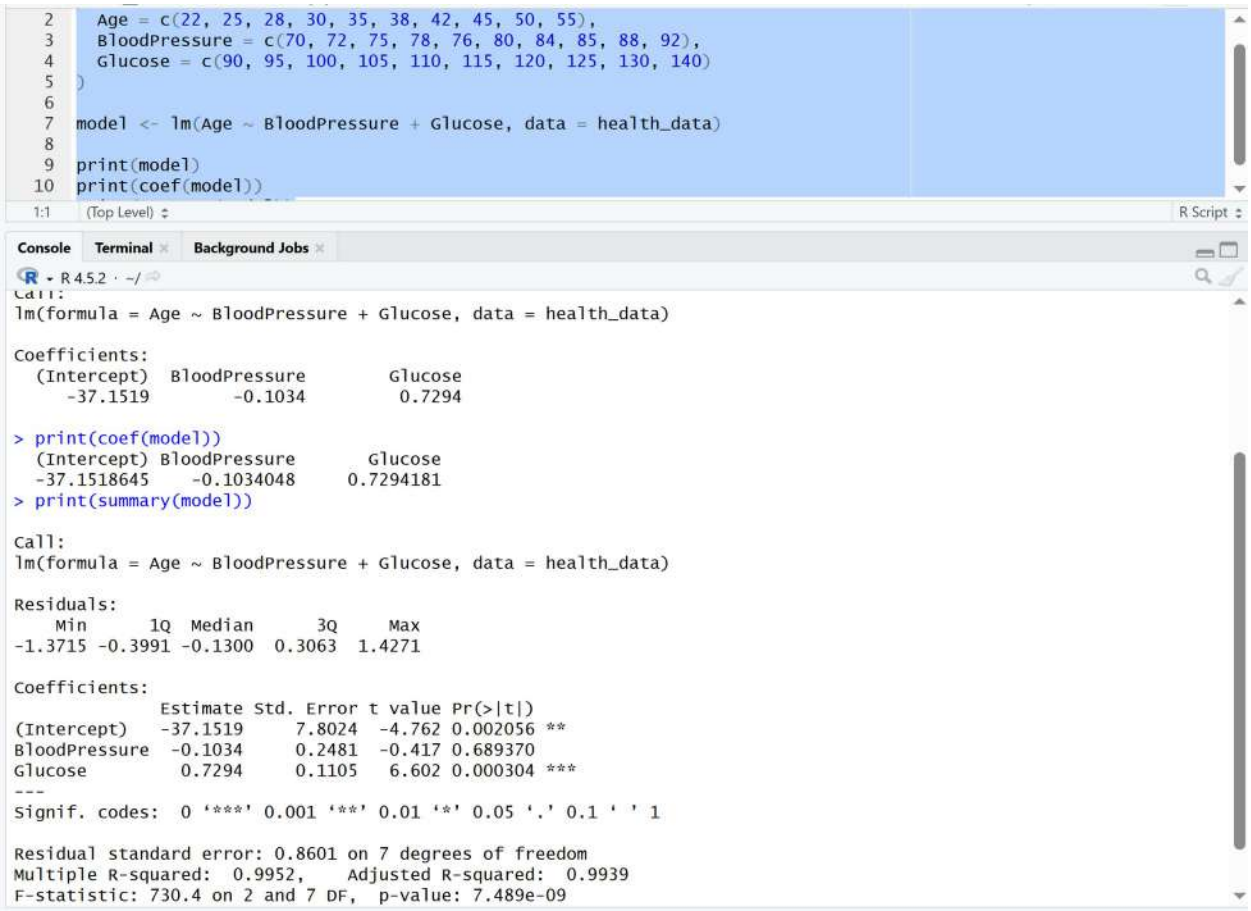
# 19. Implementing Simple Linear Regression

## Output



## 20. Implementing Multiple Linear Regression

### Output



```
  Age = c(22, 25, 28, 30, 35, 38, 42, 45, 50, 55),
  BloodPressure = c(70, 72, 75, 78, 76, 80, 84, 85, 88, 92),
  Glucose = c(90, 95, 100, 105, 110, 115, 120, 125, 130, 140)
)

model <- lm(Age ~ BloodPressure + Glucose, data = health_data)

print(model)
print(coef(model))
```

```
lm(formula = Age ~ BloodPressure + Glucose, data = health_data)

Coefficients:
  (Intercept)  BloodPressure        Glucose
     -37.1519        -0.1034         0.7294

> print(coef(model))
  (Intercept) BloodPressure       Glucose
  -37.1518645    -0.1034048     0.7294181
> print(summary(model))

Call:
lm(formula = Age ~ BloodPressure + Glucose, data = health_data)

Residuals:
    Min      1Q  Median      3Q     Max
-1.3715 -0.3991 -0.1300  0.3063  1.4271

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -37.1519     7.8024  -4.762 0.002056 **
BloodPressure  -0.1034     0.2481  -0.417 0.689370
Glucose         0.7294     0.1105   6.602 0.000304 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8601 on 7 degrees of freedom
Multiple R-squared:  0.9952,    Adjusted R-squared:  0.9939
F-statistic: 730.4 on 2 and 7 DF,  p-value: 7.489e-09
```